

41. Let a differentiable function f satisfy $f(x) + \int_3^x \frac{f(t)}{t} dt = \sqrt{x+1}, x \geq 3$. Then $12f(8)$ is equal to :

माना एक अवकलनीय फलन f , $f(x) + \int_3^x \frac{f(t)}{t} dt = \sqrt{x+1}, x \geq 3$ को संतुष्ट करता है। तो $12f(8)$ बराबर है :

- (A) 1 (B) 34 (C) 19 (D) 17

Ans. D

Sol. $f'(x) + \frac{f(x)}{x} = \frac{1}{2\sqrt{x+1}}$

$$\frac{dy}{dx} + \frac{y}{x} = \frac{1}{2\sqrt{x+1}}$$

$$\text{I.F.} = e^{\int \frac{1}{x} dx} = x$$

$$y \cdot x = \int x \frac{1}{2\sqrt{x+1}} dx + c \text{ put } x+1 = t^2, dx = 2t dt$$

$$y \cdot x = \frac{1}{2} \int \frac{(t^2-1)2t dt}{t} + c$$

$$y \cdot x = \frac{1}{2} \left(\frac{2}{3} t^3 - 2t \right) + c$$

$$y \cdot x = \frac{(\sqrt{x+1})^3}{3} - \sqrt{x+1} + c$$

$$\text{now } f(3) = 2$$

$$6 = \frac{(2)^3}{3} - 2 + c$$

$$8 - \frac{8}{3} = c$$

$$\frac{16}{3} = c$$

$$\therefore f(x) = \frac{1}{x} \left(\frac{(\sqrt{x+1})^3}{3} - (\sqrt{x+1}) + \frac{16}{3} \right)$$

$$\therefore 12f(8) = \frac{12}{8} \left(\frac{27}{3} - 3 + \frac{16}{3} \right)$$

$$= \frac{3}{2} \left(6 + \frac{16}{3} \right) = 17$$

42. The area of the region, enclosed by the circle $x^2 + y^2 = 2$ which is not common to the region bounded by the parabola $y^2 = x$ and the straight line $y = x$, is :

वृत्त $x^2 + y^2 = 2$ द्वारा परिबद्ध क्षेत्र का वह क्षेत्रफल जो परवलय, $y^2 = x$ तथा सरल रेखा, $y = x$ द्वारा परिबद्ध क्षेत्र में नहीं है, है:

- (A) $\frac{1}{6}(12\pi - 1)$ (B) $\frac{1}{6}(24\pi - 1)$ (C) $\frac{1}{3}(12\pi - 1)$ (D) $\frac{1}{3}(24\pi - 1)$

Ans. A

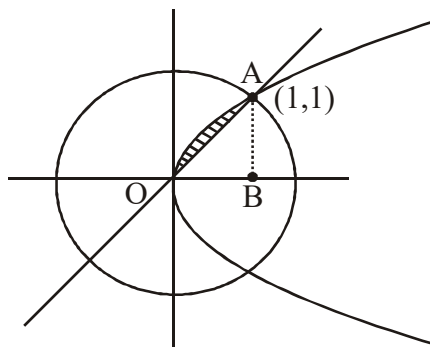
Sol. Area of circle

$$x^2 + y^2 = 2 \text{ is } \pi r^2 = 2\pi \text{ units}$$

Area bounded by $y^2 = x$ and $y = x$ is

$$A_2 = \int_0^1 (\sqrt{x} - x) dx$$

$$= \left[\frac{2x^{3/2}}{3} - \frac{x^2}{2} \right]_0^1 = \frac{1}{6}$$



So desired area

$$= A_1 - A_2 = 2\pi - \frac{1}{6} = \frac{12\pi - 1}{6} \text{ units}$$

43. The value of the integral $\int_{-2}^2 \frac{\sin^2 x}{\left[\frac{x}{\pi} \right] + \frac{1}{2}} dx$ (where $[x]$ denotes the greatest integer less than or equal to x) is :

समाकल $\int_{-2}^2 \frac{\sin^2 x}{\left[\frac{x}{\pi} \right] + \frac{1}{2}} dx$ (जहाँ $[x]$, x के समान या उससे कम महतम पूर्णांक को दर्शाता है) का मान है :

- (A) $4 - \sin 4$ (B) 4 (C) 0 (D) $\sin 4$

Ans. C

$$\begin{aligned}
 \text{S. } & \int_0^2 \left(\frac{\sin^2 x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} + \frac{\sin^2 x}{\left[\frac{-x}{\pi}\right] + \frac{1}{2}} \right) dx \\
 & \left[\frac{x}{\pi}\right] + \left[\frac{-x}{\pi}\right] = -1 \\
 & \left[\frac{-x}{\pi}\right] = -1 - \left[\frac{x}{\pi}\right] \\
 & = \int_0^2 \left(\frac{\sin^2 x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} + \frac{\sin^2 x}{-1 - \left[\frac{x}{\pi}\right] - \frac{1}{2}} \right) dx \\
 & = \int_0^2 \left(\frac{\sin^2 x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} - \frac{\sin^2 x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} \right) dx = 0
 \end{aligned}$$

44. If for all real triplets (a, b, c), $f(x) = a + bx + cx^2$, then $\int_0^1 f(x) dx$ is equal to :

यदि सभी वास्तविक त्रिकों (a, b, c), $f(x) = a + bx + cx^2$ है, तो $\int_0^1 f(x) dx$ बराबर है :

- | | |
|--|---|
| (A) $\frac{1}{2} \left\{ f(1) + 3f\left(\frac{1}{2}\right) \right\}$ | (B) $\frac{1}{6} \left\{ f(0) + f(1) + 4f\left(\frac{1}{2}\right) \right\}$ |
| (C) $2 \left\{ 3f(1) + 2f\left(\frac{1}{2}\right) \right\}$ | (D) $\frac{1}{3} \left\{ f(0) + f\left(\frac{1}{2}\right) \right\}$ |

Ans. B

Sol. $f(x) = a + bx + cx^2$

$$\begin{aligned}
 \int_0^1 f(x) dx &= \left[ax + \frac{bx^2}{2} + \frac{cx^3}{3} \right]_0^1 \\
 &= a + \frac{b}{2} + \frac{c}{3}
 \end{aligned}$$

$$f(0) = a$$

$$f(1) = a + b + c$$

$$f\left(\frac{1}{2}\right) = a + \frac{b}{2} + \frac{c}{4}$$

$$f(0) + f(1) + 4f\left(\frac{1}{2}\right) = a + (a + b + c) + (4a + 2b + c)$$

$$= 6a + 3b + 2c$$

$$\int_0^1 f(x) dx = \frac{1}{6} \{f(0) + f(1) + 4f\left(\frac{1}{2}\right)\}$$

45. The region represented by $|x - y| \leq 2$ and $|x + y| \leq 2$ is bounded by a :

(A) Square of side length $2\sqrt{2}$ units (B) Square of area 16 sq. units

(C) Rhombus of side length 2 units (D) Rhombus of area $8\sqrt{2}$ sq. units

$|x - y| \leq 2$ तथा $|x + y| \leq 2$ द्वारा प्रदर्शित क्षेत्र जिसके द्वारा प्रतिबद्ध (bounded) है, वह है :

(A) एक वर्ग जिसकी भुजा की लम्बाई $2\sqrt{2}$ इकाई है। (B) एक वर्ग जिसका क्षेत्रफल 16 वर्ग इकाई है।

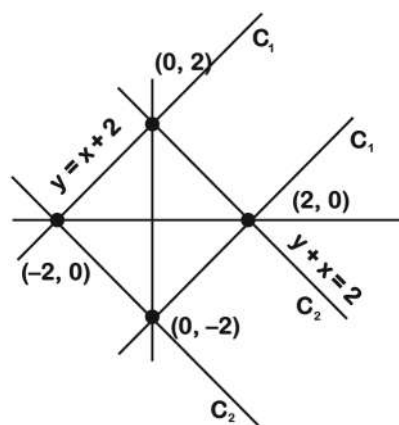
(C) एक समचतुर्भुज जिसकी भुजा की लम्बाई 2 इकाई है। (D) एक समचतुर्भुज जिसका क्षेत्रफल $8\sqrt{2}$ वर्ग इकाई है।

Ans. A

sol. $C_1 : |y - x| \leq 2$

$$C_2 : |y + x| \leq 2$$

Now region is square



$$\text{Length of side} = \sqrt{2^2 + 2^2} = 2\sqrt{2}$$

46. The integral $\int_{\pi/6}^{\pi/3} \tan^3 x \cdot \sin^2 3x (2 \sec^2 x \cdot \sin^2 3x + 3 \tan x \cdot \sin 6x) dx$ is.

समाकल $\int_{\pi/6}^{\pi/3} \tan^3 x \cdot \sin^2 3x (2 \sec^2 x \cdot \sin^2 3x + 3 \tan x \cdot \sin 6x) dx$ का मान है

- (A) $-\frac{1}{18}$ (B) $-\frac{1}{9}$ (C) $\frac{7}{18}$ (D) $\frac{9}{2}$

Ans. A

Sol.
$$I = \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \{ (4 \tan^3 x \cdot \sec^2 x) \sin^4 3x + (4 \sin^3 3x \cdot \cos 3x \cdot 3) \tan^4 x \} dx$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \{ (4 \tan^3 x \cdot \sec^2 x) \sin^4 3x + (4 \sin^3 3x \cdot \cos 3x \cdot 3) \tan^4 x \}$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin^4 3x \frac{d}{dx} (\tan^4 x) + \tan^4 x \frac{d}{dx} (\sin^4 3x) \} dx$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \left\{ \frac{d}{dx} (\tan^4 x \sin^4 3x) \right\} dx = \frac{1}{2} \left[\tan^4 x \cdot \sin^4 3x \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$$

$$= \frac{1}{2} \left[0 - 1 \frac{1}{(\sqrt{3})^4} \right] = -\frac{1}{18}$$

47. Let $x = x(y)$ be the solution of the differential equation

$2(y+2) \log_e(y+2) dx + (x+4-2 \log_e(y+2)) dy = 0, y > -1$ with $x(e^4 - 2) = 1$. Then $x(e^9 - 2)$ is equal to:

यदि $x = x(y)$ अवकल समीकरण $2(y+2) \log_e(y+2) dx + (x+4-2 \log_e(y+2)) dy = 0, y > -1$ का हल है जहाँ $x(e^4 - 2) = 1$ हो, तो $x(e^9 - 2)$ का मान बराबर होगा :

- (A) $\frac{10}{3}$ (B) $\frac{4}{9}$ (C) 3 (D) $\frac{32}{9}$

Ans. D

Sol. $2(y+2) \ln(y+2) dx + (x+4-2 \ln(y+2)) dy = 0$

$$2 \ln(y+2) + (x+4-2 \ln(y+2)) \frac{1}{y+2} \frac{dy}{dx} = 0$$

$$\text{let } \ln(y+2) = t$$

$$\frac{1}{y+2} \cdot \frac{dy}{dx} = \frac{dt}{dx}$$

$$2t + (x + 4 - 2t) \cdot \frac{dt}{dx} = 0$$

$$(x + 4 - 2t) \frac{dt}{dx} = -2t$$

$$\frac{dx}{dt} = \frac{2t - 4 - x}{2t}$$

$$\frac{dx}{dt} + \frac{x}{2t} = \frac{2t - 4}{2t}$$

$$x \cdot t^{1/2} = \int \frac{2t - 4}{2t} \cdot t^{1/2} \cdot dt$$

$$x \cdot t^{1/2} = \int \left(t^{1/2} - \frac{2}{t^{1/2}} \right) \cdot dt$$

$$x t^{\frac{1}{2}} = \frac{t^{\frac{3}{2}}}{\frac{3}{2}} - 2 \cdot \frac{t^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$x \cdot t^{\frac{1}{2}} = \frac{2t^{\frac{3}{2}}}{3} - 4t^{\frac{1}{2}} + C$$

$$x = \frac{2}{3} \cdot t - 4 + C t^{-\frac{1}{2}}$$

$$x = \frac{2}{3} \ln(y + 2) - 4 + C (\ln(y + 2))^{-\frac{1}{2}}$$

$$\text{Put } y = e^4 - 2, x = 1$$

$$1 = \frac{2}{3} \times 4 - 4 + C \times \frac{1}{2}$$

$$\frac{C}{2} = 5 - \frac{8}{3} = \frac{7}{3}$$

$$\Rightarrow C = \frac{14}{3}$$

$$\begin{aligned}
 x &= \frac{2}{3} \times 9 - 4 + \frac{14}{3} \times \frac{1}{3} \\
 &= 2 + \frac{14}{9} \\
 &= \frac{32}{9}
 \end{aligned}$$

48. Which of the following statements is incorrect for the function $g(\alpha)$ for $\alpha \in \mathbb{R}$ such that

$$g(\alpha) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^\alpha x}{\cos^\alpha x + \sin^\alpha x} dx$$

- (A) $g(\alpha)$ is an even function
- (B) $g(\alpha)$ is a strictly increasing function
- (C) $g(\alpha)$ has an inflection point at $\alpha = -\frac{1}{2}$
- (D) $g(\alpha)$ is a strictly decreasing function

फलन $g(\alpha) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^\alpha x}{\cos^\alpha x + \sin^\alpha x} dx$, $\alpha \in \mathbb{R}$ के लिए निम्न में से कौनसा कथन असत्य है ?

- (A) $g(\alpha)$ एक सम फलन है।
- (B) $g(\alpha)$ एक निरंतर वर्धमान फलन है।
- (C) $g(\alpha)$ का $\alpha = -\frac{1}{2}$ पर एक नति परिवर्तन बिन्दु है।
- (D) $g(\alpha)$ एक निरंतर ह्रासमान फलन है।

Ans. A

Sol. $g(\alpha) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{(\sin x)^\alpha}{(\sin x)^\alpha + (\cos x)^\alpha} dx \dots\dots(1)$

apply king

$$g(\alpha) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{(\cos x)^\alpha}{(\cos x)^\alpha + (\sin x)^\alpha} dx \quad \dots\dots\dots(2)$$

add (1) + (2)

$$2g(\alpha) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} dx$$

$$g(\alpha) = \frac{\pi}{12}$$

49. The solution of the differential equation $\frac{dy}{dx} - \frac{y+3x}{\log_e(y+3x)} + 3 = 0$ is : (where C is a constant of integration.)

अवकल समीकरण $\frac{dy}{dx} - \frac{y+3x}{\log_e(y+3x)} + 3 = 0$ का हल है : (जहाँ C एक समाकलन अचर है।)

(A) $x - 2\log_e(y+3x) = C$

(B) $x - \frac{1}{2}(\log_e(y+3x))^2 = C$

(C) $x - \log_e(y+3x) = C$

(D) $y + 3x - \frac{1}{2}(\log_e x)^2 = C$

Ans. B

Sol. $\frac{dy}{dx} - \frac{y-3x}{\ln(y-3x)} = 3$

$$y - 3x = t$$

$$\frac{dy}{dx} - 3 = \frac{dt}{dx}$$

$$\frac{dt}{dx} + 3 - \frac{t}{\ln(t)} = 3$$

$$\int \frac{\ln(t) dx}{t} = \int dx$$

$$\ln(t) = p$$

$$\frac{1}{t} dt = dp$$

$$\int p dp = \int dx$$

$$\frac{p^2}{2} = x + c$$

$$\frac{\ln^2(t)}{2} = x + c$$

$$\frac{\ln^2(y-3x)}{2} = x + c$$

50. The value of $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{(2+3\sin x)}{\sin x(1+\cos x)} dx$ is equal to

$$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{(2+3\sin x)}{\sin x(1+\cos x)} dx \text{ का मान बराबर है}$$

(A) $\frac{10}{3} - \sqrt{3} - \log_e \sqrt{3}$

(B) $-2 + 3\sqrt{3} + \log_e \sqrt{3}$

(C) $\frac{10}{3} - \sqrt{3} + \log_e \sqrt{3}$

(D) $\frac{7}{2} - \sqrt{3} - \log_e \sqrt{3}$

Ans. C

Sol. $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{2(1-\cos x)}{\sin x(1-\cos^2 x)} dx + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{3}{1+\cos x} dx$

$$= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{2}{\sin^3 x} dx - 2 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cot x \cdot \operatorname{cosec}^2 x dx + \frac{3}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sec^2 \frac{x}{2} dx$$

$$= 2 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sqrt{1+\cot^2 x} \cdot \operatorname{cosec}^2 x dx - 2 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cot x \cdot \operatorname{cosec}^2 x dx + 3 \left[\tan \frac{x}{2} \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

Let $\cot x = 1$

$$\begin{aligned}
 &= -2 \int_{\frac{1}{\sqrt{3}}}^0 \sqrt{1+t^2} dt + 2 \int_{\frac{1}{\sqrt{3}}}^0 t dt + 3 \left[1 - \frac{1}{\sqrt{3}} \right] \\
 &= 2 \left[\frac{t}{2} \sqrt{1+t^2} + \frac{1}{2} \log(t + \sqrt{1+t^2}) \right]_{\frac{1}{\sqrt{3}}}^0 - \left[t^2 \right]_{\frac{1}{\sqrt{3}}}^0 + 3 - \sqrt{3} \\
 &= \frac{2}{3} + \log \sqrt{3} - \frac{1}{3} + 3 - \sqrt{3} = \frac{10}{3} - \sqrt{3} + \frac{1}{2} \log 3
 \end{aligned}$$

51. If $[t]$ denotes the greatest integer $\leq t$, then the value of $\frac{3(e-1)}{e} \int_1^2 x^2 e^{[x]+[x^3]} dx$ is :

यदि $[t]$ महत्तम पूर्णांक $\leq t$, $\frac{3(e-1)}{e} \int_1^2 x^2 e^{[x]+[x^3]} dx$ का मान है :

- (A) $e^8 - 1$ (B) $e^7 - 1$ (C) $e^8 - e$ (D) $e^9 - 1$

Ans. C

Sol. $\frac{3(e-1)}{e} \int_1^2 x^2 e^{[x]+[x^3]} dx$

Let $k = \int_1^2 x^2 e^{1+[x^3]} dx$

$x^3 = t$

$3x^2 dx = dt$

$k = \frac{1}{3} \int_1^8 e \cdot e^{[t]} dt$

$\frac{e}{3} [e + e^2 + e^3 + \dots + e^7] = \frac{e^2}{3} \left(\frac{e^7 - 1}{e - 1} \right)$

$\frac{3(e-1)}{e} k = \frac{3(e-1)}{e} \frac{e^2}{3} \left(\frac{e^7 - 1}{e - 1} \right) = e(e^7 - 1)$

$= e^8 - e$

52. $\int \frac{\sin \frac{5x}{2}}{\sin \frac{x}{2}} dx$ is equal to (where c is a constant of integration)

$\int \frac{\sin \frac{5x}{2}}{\sin \frac{x}{2}} dx$ बराबर है : (जहाँ c एक समाकलन अचर है।)

(A) $2x + \sin x + 2\sin 2x + c$

(B) $x + 2 \sin x + 2\sin 2x + c$

(C) $x + 2 \sin x + \sin 2x + c$

(D) $2x + \sin x + \sin 2x + c$

Ans. C

sol. $\int \frac{\sin\left(\frac{5x}{2}\right)}{\sin\left(\frac{x}{2}\right)} dx$

$$= \int \frac{2 \cos \frac{x}{2} \cdot \sin \frac{5x}{2}}{2 \cos \frac{x}{2} \cdot \sin \frac{x}{2}} dx$$

$$= \int \frac{\sin 3x + \sin 2x}{\sin x} dx$$

Now use $\sin 2x = 2 \sin x \cos x$ and $\sin 3x = 3 \sin x - 4 \sin^3 x$

$$= \int (1 + 2 \cos x + 2 \cos 2x) dx$$

$$= x + 2 \sin x + \sin 2x + c$$

53. The general solution of the differential equation $(y^2 - x^3) dx - xy dy = 0$ ($x \neq 0$) is :

(where c is a constant of integration)

अवकल समीकरण $(y^2 - x^3) dx - xy dy = 0$ ($x \neq 0$) का व्यापक हल है :

(जहाँ c एक समाकलन अचर है)

(A) $y^2 + 2x^3 + cx^2 = 0$

(B) $y^2 - 2x^2 + cx^3 = 0$

(C) $y^2 - 2x^3 + cx^2 = 0$

(D) $y^2 + 2x^2 + cx^3 = 0$

Ans. A

sol. $y^2 dx - xy dy = x^3 dx$

$$\Rightarrow \frac{(y dx - x dy) y}{x^2} = x dx$$

$$\Rightarrow -y d\left(\frac{y}{x}\right) = x dx$$

$$\Rightarrow -\frac{y}{x} \cdot d\left(\frac{y}{x}\right) = dx$$

$$\Rightarrow -\frac{1}{2} \left(\frac{y}{x}\right)^2 = x + c_1$$

$$\Rightarrow 2x^3 + cx^2 + y^2 = 0$$

54. If $x^3 dy + xy dx = x^2 dy + 2y dx$; $y(2) = e$ and $x > 1$, then $y(4)$ is equal to :

यदि $x^3 dy + xy dx = x^2 dy + 2y dx$; $y(2) = e$ तथा $x > 1$, तो $y(4)$ बराबर है :

(A) $\frac{\sqrt{e}}{2}$ (B) $\frac{1}{2} + \sqrt{e}$ (C) $\frac{3}{2}\sqrt{e}$ (D) $\frac{3}{2} + \sqrt{e}$

Ans. C

Sol. $x^3 dy + xy dx = 2y dx + x^2 dy$

By variable seprable method

$$\Rightarrow (x^3 - x^2)dy = (2 - x)ydx$$

$$\Rightarrow \frac{dy}{y} = \frac{2 - x}{x^2(x - 1)} dx$$

$$\Rightarrow \int \frac{dy}{y} = \int \frac{2 - x}{x^2(x - 1)} dx$$

By partial fraction

Let $\frac{2 - x}{x^2(x - 1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x - 1}$

$$\Rightarrow 2 - x = Ax(x - 1) + B(x - 1) + Cx^2$$

By comparison

$$\Rightarrow C = 1, B = -2 \text{ and } A = -1$$

$$\Rightarrow \int \frac{dy}{y} = \int \left\{ \frac{-1}{x} - \frac{2}{x^2} + \frac{1}{x - 1} \right\} dx$$

$$\Rightarrow \ln y = -\ln x + \frac{2}{x} + \ln |x - 1| + C$$

$$\therefore y(2) = e$$

$$\Rightarrow \ln e = -\ln 2 + \frac{2}{2} + \ln 1 + C$$

$$\Rightarrow C = \ln 2$$

$$\Rightarrow \ln y = -\ln x + \frac{2}{x} + \ln |x-1| + \ln 2$$

$$\text{at } x = 4$$

$$\Rightarrow \ln y(4) = -\ln 4 + \frac{1}{2} + \ln 3 + \ln 2$$

$$\Rightarrow \ln y(4) = \ln \left(\frac{3}{2} \right) + \frac{1}{2} = \ln \left(\frac{3}{2} e^{\frac{1}{2}} \right)$$

$$\Rightarrow y(4) = \frac{3}{2} e^{\frac{1}{2}}$$

55. The area (in sq. units) of the region $A = \{(x, y) : (x-1)[x] \leq y \leq 2\sqrt{x}, 0 \leq x \leq 2\}$, Where $[t]$ denotes the greatest integer function, is :

क्षेत्र $A = \{(x, y) : (x-1)[x] \leq y \leq 2\sqrt{x}, 0 \leq x \leq 2\}$, जहाँ $[t]$ महत्तम पूर्णांक फलन है, का क्षेत्रफल (वर्ग इकाइयों में) है:

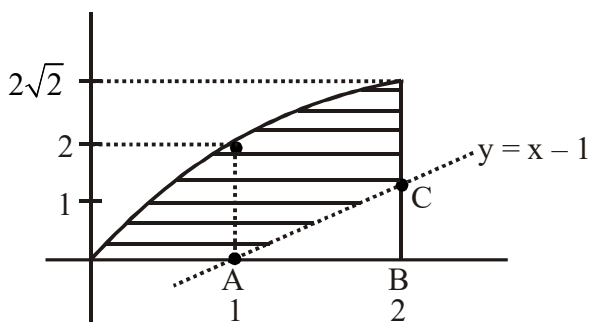
- (A) $\frac{8}{3}\sqrt{2}-1$ (B) $\frac{4}{3}\sqrt{2}+1$ (C) $\frac{4}{3}\sqrt{2}-\frac{1}{2}$ (D) $\frac{8}{3}\sqrt{2}-\frac{1}{2}$

Ans. D

Sol. Sol. $[x](x-1) \leq y \leq 2\sqrt{x}$

$$0(x-1) \leq y \leq 2\sqrt{x} \Rightarrow 0 \leq y \leq 2\sqrt{x} \quad 0 \leq x < 1$$

$$(1)(x-1) \leq y \leq 2\sqrt{x} \quad 1 \leq x < 2$$



$$\text{Required Area} = \int_0^2 2\sqrt{x} dx - \frac{1}{2}(1)(1)$$

$$= 2 \cdot \left(\frac{(x)^{3/2}}{3/2} \right)_0^2 - \frac{1}{2}$$

$$= \frac{4}{3} [2\sqrt{2}] - \frac{1}{2}$$

$$= \frac{8\sqrt{2}}{3} - \frac{1}{2}$$

56. If $\int \frac{x+1}{\sqrt{2x-1}} dx = f(x)\sqrt{2x-1} + C$, where C is a constant of integration, then f(x) is equal to :

यदि $\int \frac{x+1}{\sqrt{2x-1}} dx = f(x)\sqrt{2x-1} + C$ है, जहाँ C एक समाकलन अचर है, तो f(x) बराबर है -

- (A) $\frac{2}{3}(x+2)$ (B) $\frac{2}{3}(x-4)$ (C) $\frac{1}{3}(x+1)$ (D) $\frac{1}{3}(x+4)$

Ans. D

Sol. $\int \frac{(x+1)}{\sqrt{2x-1}} dx = f(x)\sqrt{2x-1} + c$

$$2x-1 = t^2$$

$$2dx = 2tdt \Rightarrow dx = tdt$$

$$x = \frac{t^2+1}{2}$$

$$\int \frac{t^2+3}{2 \times t} \cdot tdt = \frac{1}{2} \int t^2 dt + \frac{3}{2} \int dt$$

$$= \frac{1}{2} \cdot \frac{t^3}{3} + \frac{3}{2} t + c$$

$$= \frac{1}{6} (2x-1)^{3/2} + \frac{3}{2} (2x-1)^{1/2} + c$$

$$= (\sqrt{2x-1}) \left[\frac{2x-1}{6} + \frac{3}{2} \right] + c$$

$$= (\sqrt{2x-1}) \left[\frac{2x-1+9}{6} \right] + c$$

$$= (\sqrt{2x-1}) \left(\frac{x+4}{3} \right) + c$$

57. If $\int e^{\sec x} (\sec x \tan x f(x) + \sec x \tan x + \sec^2 x) dx = e^{\sec x} f(x) + C$, then a possible choice of $f(x)$ is :

यदि $\int e^{\sec x} (\sec x \tan x f(x) + \sec x \tan x + \sec^2 x) dx = e^{\sec x} f(x) + C$, तो $f(x)$ का एक संभव विकल्प है :

(A) $\sec x - \tan x - \frac{1}{2}$ (B) $\sec x + \tan x + \frac{1}{2}$ (C) $\sec x + x \tan x - \frac{1}{2}$ (D) $x \sec x + \tan x + \frac{1}{2}$

Ans. B

sol. $\int e^{\sec x} (\sec x \tan x f(x) + (\sec x \tan x + \sec^2 x)) dx = e^{\sec x} f(x) + C$

$\therefore$ We know that

$$\int e^{g(x)} ((g'(x)f(x)) + f'(x)) dx = e^{g(x)} \times f(x) + C$$

$$\therefore f(x) = \int ((\sec x \tan x) + \sec^2 x) dx$$

$$\therefore \boxed{f(x) = \sec x + \tan x + C}$$

58. The value of the integral $\int_{1/2}^2 \frac{\tan^{-1} x}{x} dx$ is equal to

समाकलन $\int_{1/2}^2 \frac{\tan^{-1} x}{x} dx$ का मान बराबर है

(A) $\pi \log_e 2$ (B) $\frac{1}{2} \log_e 2$ (C) $\frac{\pi}{4} \log_e 2$ (D) $\frac{\pi}{2} \log_e 2$

Ans. D

Sol. $I = \int_{1/2}^2 \frac{\tan^{-1} x}{x} dx$ (i)

Put $x = \frac{1}{t}$ $dx = -\frac{1}{t^2} dt$

$$I = - \int_2^{1/2} \frac{\tan^{-1} \frac{1}{t}}{\frac{1}{t}} \cdot \frac{1}{t^2} dt = - \int_2^{1/2} \frac{\tan^{-1} \frac{1}{t}}{t} dt$$

$$I = \int_2^{1/2} \frac{\cot^{-1} t}{t} dt = \int_2^{1/2} \frac{\cot^{-1} x}{x} dx \quad \dots(ii)$$

Add both equation

$$2I = \int_{1/2}^2 \frac{\tan^{-1} x + \cot^{-1} x}{x} dx = \frac{\pi}{2} \int_{1/2}^2 \frac{dx}{x} = \frac{\pi}{2} (\ln 2)^2_{1/2}$$

$$= \frac{\pi}{2} \left(\ln 2 - \ln \frac{1}{2} \right) = \pi \ln 2$$

$$I = \frac{\pi}{2} \ln 2$$

59. The solution of the differential equation $\frac{dy}{dx} = -\left(\frac{x^2 + 3y^2}{3x^2 + y^2}\right)$, $y(1) = 0$ is

अवकल समीकरण $\frac{dy}{dx} = -\left(\frac{x^2 + 3y^2}{3x^2 + y^2}\right)$, $y(1) = 0$ का हल है :

(A) $\log_e |x + y| - \frac{xy}{(x + y)^2} = 0$

(B) $\log_e |x + y| + \frac{xy}{(x + y)^2} = 0$

(C) $\log_e |x + y| + \frac{2xy}{(x + y)^2} = 0$

(D) $\log_e |x + y| - \frac{2xy}{(x + y)^2} = 0$

Ans. C

Sol. Let $y = vx$

$$v + x \frac{dv}{dx} = -\left(\frac{1 + 3v^2}{3 + v^2}\right)$$

$$\Rightarrow \frac{v^2 + 3}{(v + 1)^3} dv + \frac{1}{x} dx = 0$$

$$\Rightarrow \int \frac{4}{(v + 1)^3} dv + \int \frac{1}{v + 1} dv - \int \frac{2}{(v + 1)^2} dv + \int \frac{1}{x} dx = 0$$

$$\Rightarrow \frac{-2}{(v + 1)^2} + \ln|v + 1| + \frac{2}{v + 1} + \ln|x| = c$$

$$\Rightarrow \frac{-2x^2}{(x+y)^2} + \ln \left| \frac{x+y}{x} \right| + \frac{2x}{x+y} + \ln |x| = c$$

$$\Rightarrow \frac{2xy}{(x+y)^2} + \ln |x+y| = c$$

$$\because y(1) = 0$$

$$\Rightarrow c = 0$$

$$\Rightarrow \frac{2xy}{(x+y)^2} + \ln |x+y| = 0$$

60. The integral $\int_1^2 e^x \cdot x^x (2 + \log_e x) dx$ equals :

समाकल $\int_1^2 e^x \cdot x^x (2 + \log_e x) dx$ बराबर है :

(A) $4e^2 - 1$ (B) $e(2e - 1)$ (C) $e(4e + 1)$ (D) $e(4e - 1)$

Ans. D

Sol. $\int_1^2 e^x \cdot x^x (2 + \log_e x) dx$

$$\int_1^2 e^x (2x^x + x^x \log_e x) dx$$

$$\int_1^2 e^x (x^x + x^x (1 + \log_e x)) dx$$

$$\int e^x (f(x) + f'(x)) dx = e^x f(x) + c$$

$$\Rightarrow e^x x^x \Big|_1^2$$

$$= e^2 \times 4 - e = 4e^2 - e$$